



Faculty of Computer Studies and Information Technology
Department of Information Technology
IT 303 / IT 303L – SYSTEM INTEGRATION AND ARCHITECTURE 1
LABORATORY ACTIVITY 2 — First Semester, AY 2026-2027

SHARED VERSION — every student does the same work with the same values. If you meant to issue the individual version, it is in Individual Activities\.

Activity 2: Measurement table, sizing justification, loop still responding

Intended Learning Outcome

ILO 2: Measure the circuit and size its components so that every part still behaves as predicted once it is in the system.

Time allotment: 3 hours — the second laboratory session for this outcome. The first session covers the taught work, so ILO 2's 6 hours in the syllabus is the two sessions together.

Objectives

- Connect a meter correctly for voltage, current and resistance.
- Predict a value from Ohm's Law and write the prediction down BEFORE measuring.
- Account for the difference between predicted and measured values.
- Resize a component from the numbers, and show the loop still responding afterwards.

Materials

- Your working circuit from Activity 1.
- A prediction table drawn before you start — quantity, predicted, measured, difference.
- Tool for this activity: Tinkercad Circuits, with the simulated multimeter.

Procedure

1. Write your four predictions: supply voltage, voltage across the LED, voltage across the resistor, and current in that branch. Use $V = I \times R$ and the LED forward drop of about 2.0 V. Show the table to your instructor before continuing.
2. Place the multimeter. Set it to volts and connect it ACROSS the supply rails. Record the reading.
3. Move it across the LED, still in volts. Record.
4. Move it across the resistor. Record. Check that this reading plus the LED reading is close to the supply.
5. Set it to amps. Break the branch and insert the meter IN SERIES, so the current flows through it. Record.
6. Fill in the difference column as a percentage for each row.
7. Write one sentence accounting for the differences. A good account explains several rows with one cause.
8. Now resize: choose a resistor that gives about 15 mA instead. Compute the value first, pick the nearest standard value, and say why you picked that one.
9. Change the resistor, then start the simulation and demonstrate the loop again — press and release, LED and serial line both following.



10. Annotate your integration map with the measured values on the junction they belong to.

What to Submit

- The completed prediction table with all four rows and the difference column.

quantity	predicted	measured	difference
supply	5.00 V	5.00 V	0.0%
across LED	2.00 V	1.62 V	-19.0%
across resistor	3.00 V	2.45 V	-18.3%
current in branch	13.6 mA	11.1mA	-18.38%

- Your one-sentence account of the differences.

The measured LED forward drop is lower than the predicted value (1.62V vs. 2.00V), and because component tolerances reduced the overall voltage and current through the branch by approximately 18%, the circuit operates consistently at a lower overall power level than predicted.

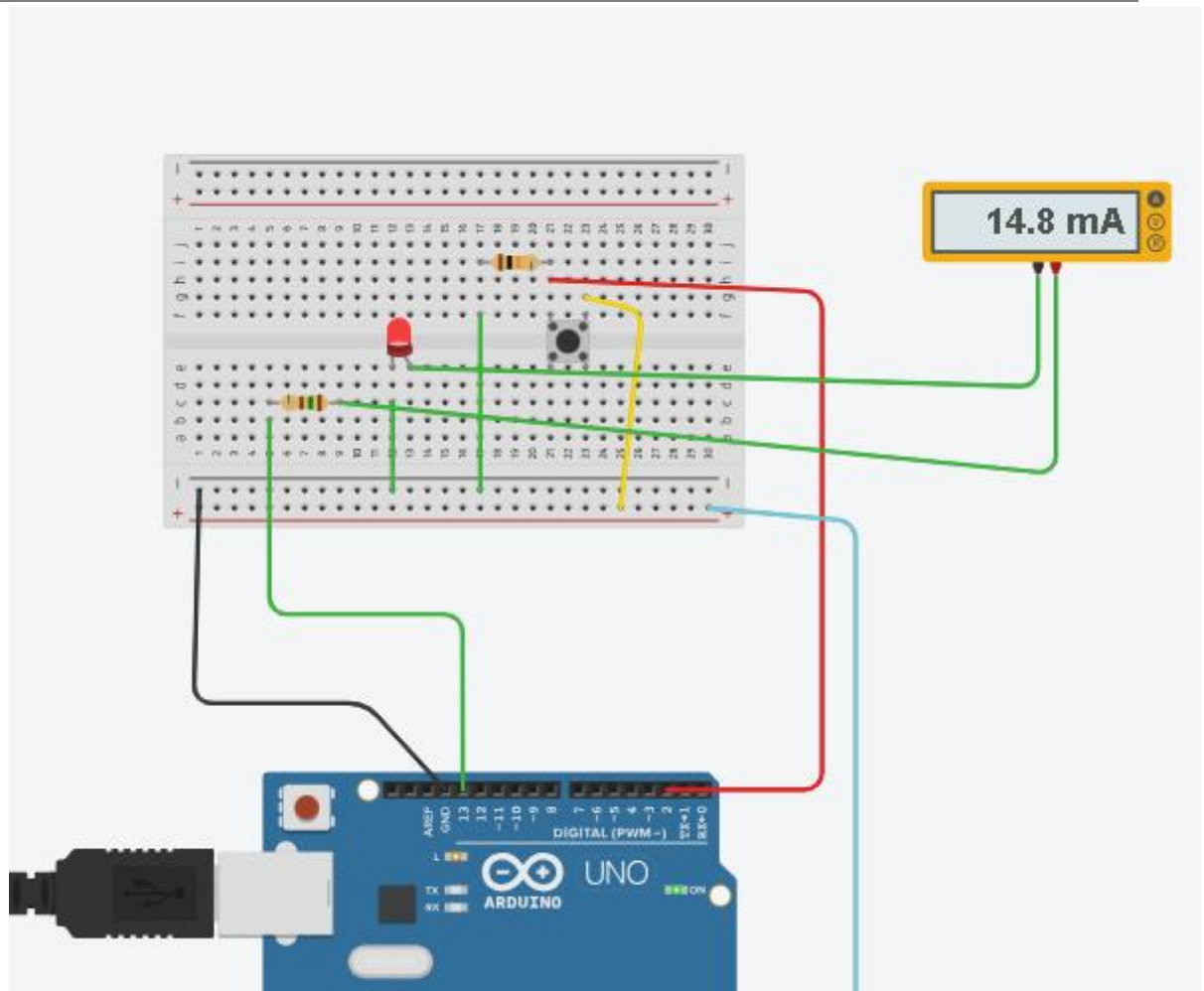
- The resize calculation, the value chosen and why.

1. Formula:

$$R = \frac{V_{\text{supply}} - V_{\text{LED}}}{I} = \frac{5.00 \text{ V} - 1.62 \text{ V}}{0.015 \text{ A}} = \frac{3.38 \text{ V}}{0.015 \text{ A}} = 225.33 \Omega$$

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- A screenshot of the loop working after the change.



How This Activity Is Marked

Rubric For Laboratory Activities, from the course syllabus. Each criterion is rated Excellent (4), Very Satisfactory (3), Satisfactory (2) or Needs Improvement (1), and that share of its weight is awarded.

Criterion	Weight	What earns full marks here
Working System	30%	The system responds from end to end at the close of the session, and the part added this session works within it.
Program Quality	25%	The program is correct, readable and uses functions and variables appropriately.
Interface Note and Evidence	20%	Every junction made this session has an interface note giving signal, units, range, timing and absence behaviour, and the recorded readings support it.
Explanation	15%	Explains what each part does and why, including any fault encountered.
Timeliness and Compliance with Instructions	10%	Submitted within the session and follows every instruction given.

Reminders

- Everything runs in a browser on the laboratory computers. You buy nothing, and you install nothing.
- Save your own copy of every file. Work that exists only on a laboratory computer is not backed up.
- Evidence means a screenshot showing the thing AND the state you are claiming — the circuit running, the value readable, the serial line visible.
- Every session ends the same way: the system responds from end to end, and you have demonstrated it. A session that ends with a system that does not run is not finished, whatever else was built.
- Every junction you make gets an interface note, written at the bench while you make it. A note written later from memory records what you believe, not what is true.
- Your device code identifies your work. It is an opaque code and it is the only identifier that belongs in a project — never your name or your student number.



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